Week 2 – Client Workstation & Domain Join

AccessGuard Health – Healthcare IAM Operations Simulation

Lab record: building HVMC-CLIENT01, joining it to HVMC.LAN and signing in as a domain user

Project Name	AccessGuard Health
Project Type	Healthcare IAM Operations Simulation – Home Lab
Organization	Horizon Valley Medical Center (HVMC) – fictional
Project Phase	Client workstation build and Active Directory domain join
Document Version	1.0
Prepared By	Sabitha S Karanavar
Lab Build Date	November 1, 2026

1. Purpose

This document records the build of the hospital workstation **HVMC-CLIENT01** and shows, step by step with screenshots, how it was joined to the **HVMC.LAN** Active Directory domain created in Week 1. It ends with a hospital employee, **sarah.chen**, signing in to the workstation with a domain account.

Everything here is a fictional training environment: no real patient data or real credentials are used.

2. Lab Environment

Component	Domain controller	Client workstation
Name	hvmc-dc01	hvmc-client01
Address	192.168.56.10 (fixed)	192.168.56.11 (fixed)
Role	Samba4 Active Directory domain controller, DNS, Kerberos	Domain member (joined with realmd and sssd)
OS / hypervisor	Ubuntu Server 24.04.5 on VirtualBox 7.0.26	Ubuntu Server 24.04.5 on VirtualBox 7.0.26
Resources	20 GB disk	2 GB memory, 1 CPU, 20 GB disk
DNS server	itself (192.168.56.10)	192.168.56.10 (the domain controller)
Networks	NAT (internet) + Host-only HostNetwork	NAT (internet) + Host-only HostNetwork

2.1 How this builds on the domain controller

Built on the domain controller (Week 1)	Used by the client
Fixed address 192.168.56.10 and Samba DNS (hvmc.lan zone)	The client's DNS server; lets it find hvmc-dc01.hvmc.lan and discover the domain
Kerberos realm HVMC.LAN	The secure login system the client joins
Administrator account (password expiry disabled)	Approved the domain join
OU=Computers	Where the client's computer account was created
10 department OUs, 10 security groups, 30 users	The employees who can sign in, such as sarah.chen in SG-Doctors-EHR-RW
Host-only network HostNetwork	The private network both VMs share
Snapshots 01–03	Safety net for the controller; the client now has its own snapshots 01 and 02

3. Client Virtual Machine Creation

The VM **HVMC-CLIENT01** was created in VirtualBox with 2 GB memory, 1 processor and a 20 GB disk. **Why small?** The 2014 MacBook Air has 8 GB of memory, and the domain controller must run at the same time. **Why two network adapters?** NAT gives internet access for updates; the host-only adapter puts the client on the private lab network with the domain controller.

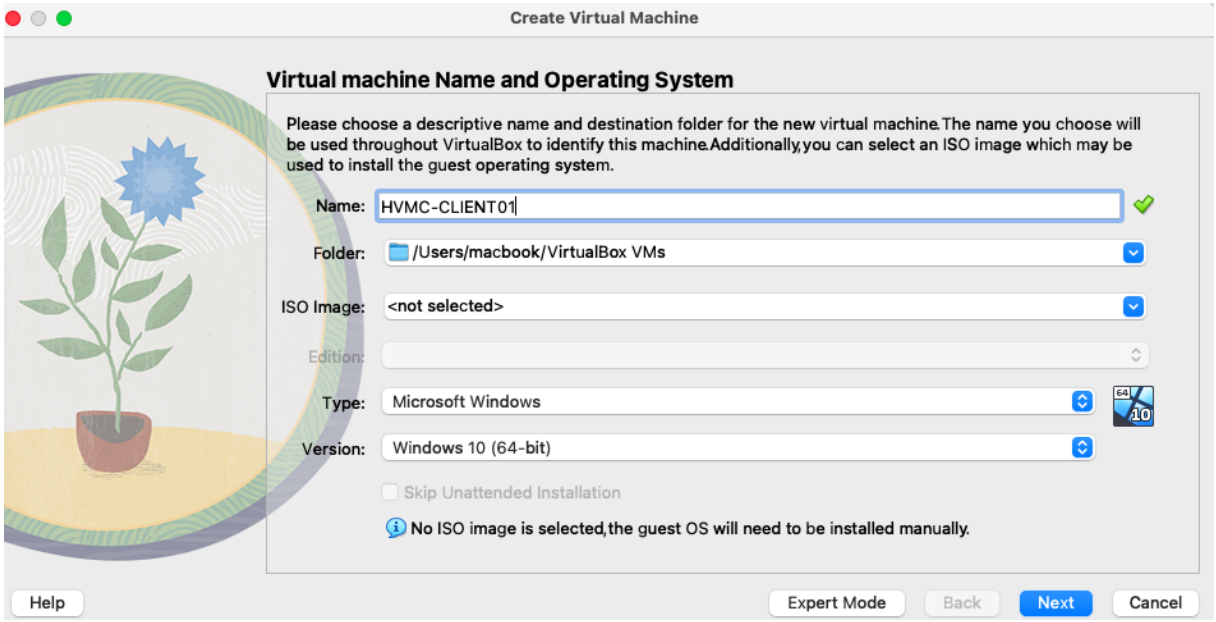


Figure 1: New virtual machine named HVMC-CLIENT01

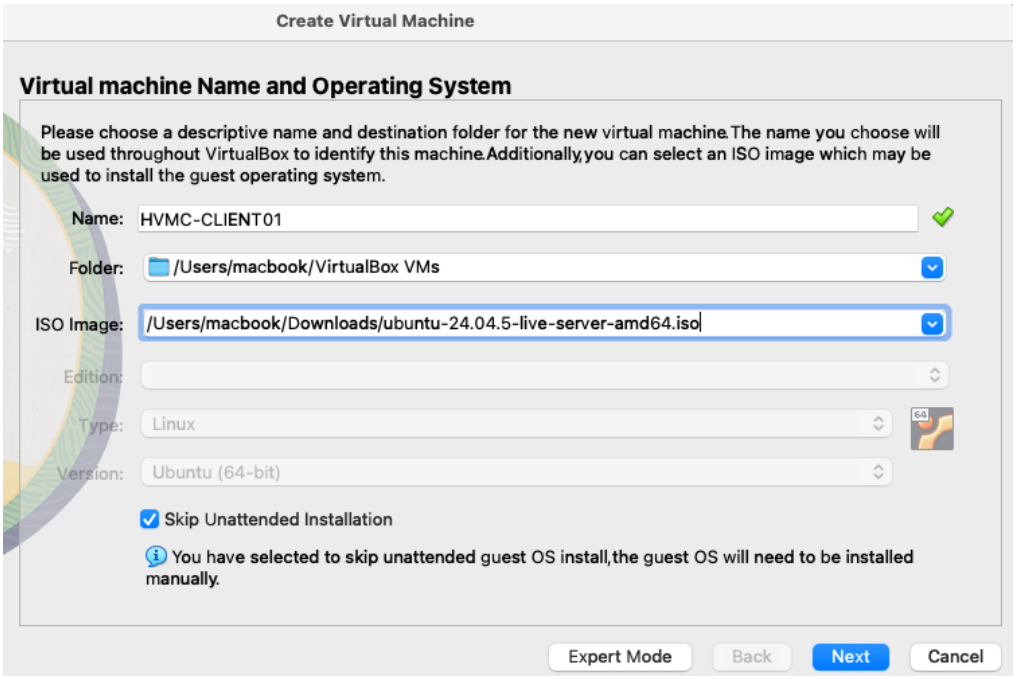


Figure 2: Ubuntu Server 24.04.5 ISO selected; unattended install skipped

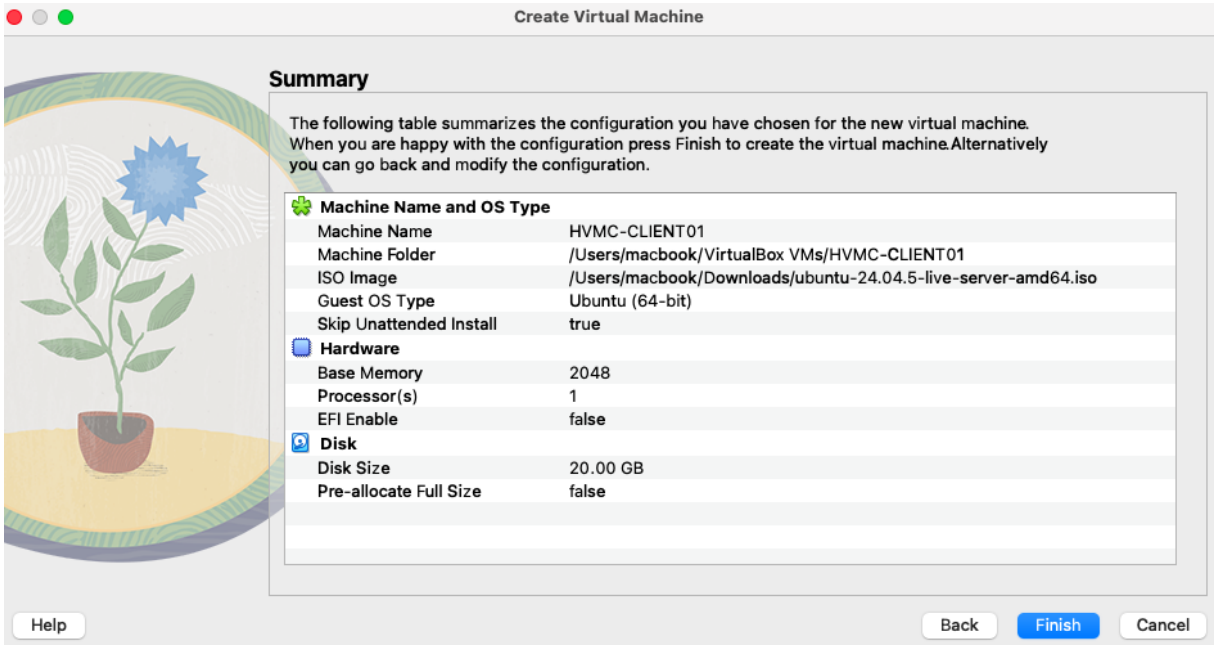


Figure 3: Summary: 2048 MB memory, 1 processor, 20 GB disk

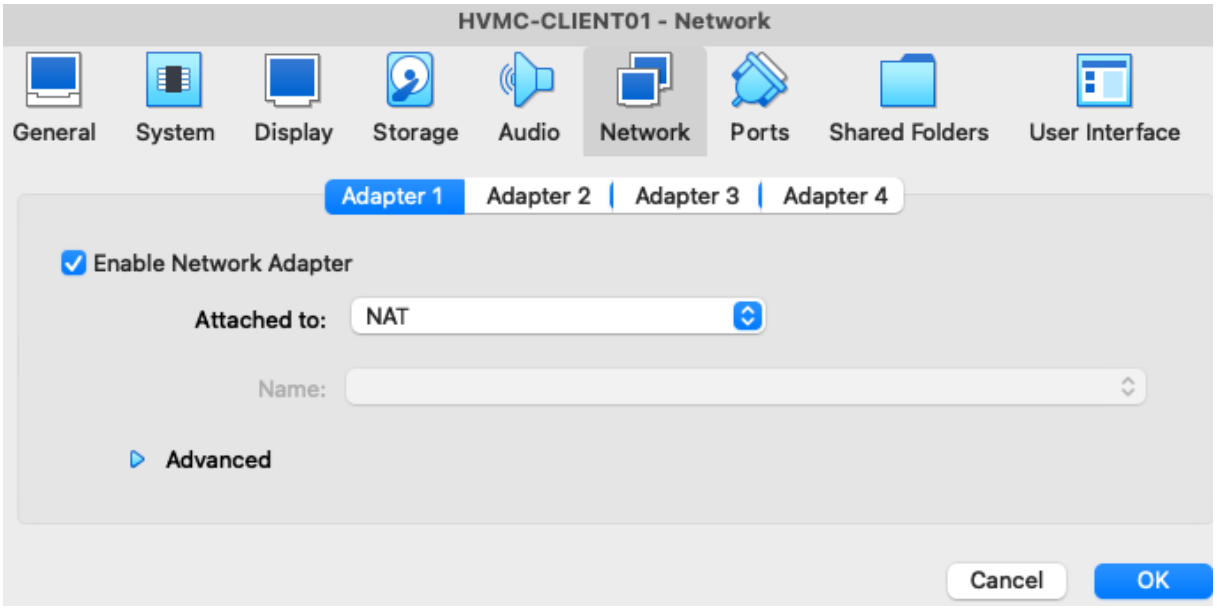


Figure 4: Adapter 1 – NAT

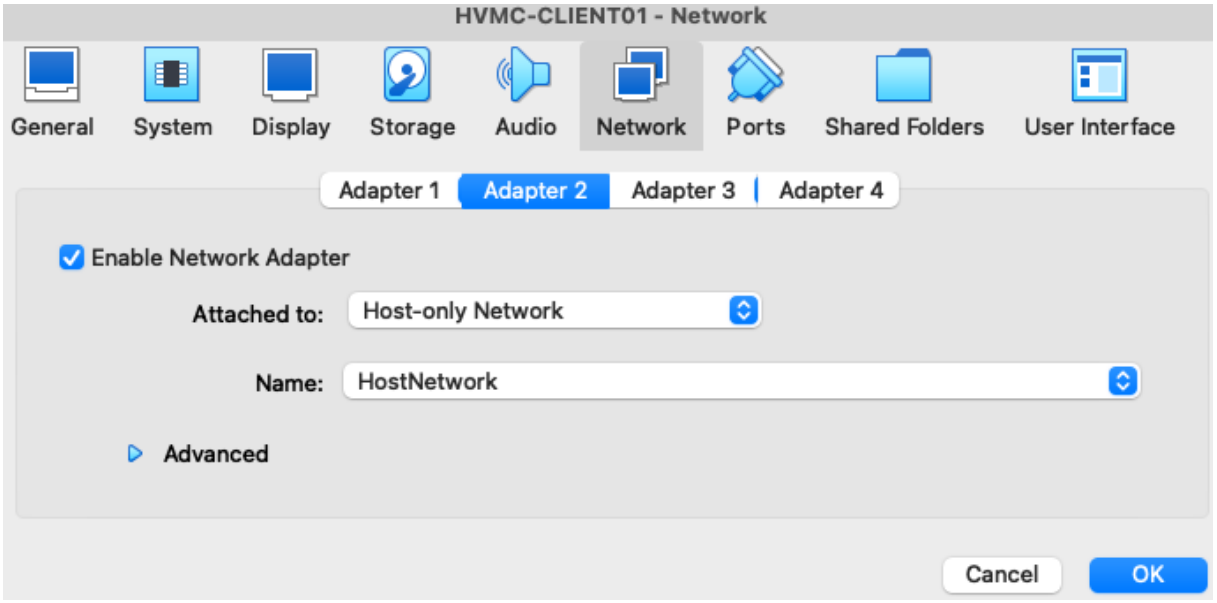


Figure 5: Adapter 2 – Host-only Network (HostNetwork)

4. Ubuntu Server Installation

The installer defaults were accepted for keyboard, server type, network, proxy and mirror. The whole 20 GB disk was used with LVM, the server name was set to **hvmc-client01**, and OpenSSH was installed so the VM can be managed from the Mac Terminal. The first attempt was rebuilt after a password problem (see the troubleshooting log), so the install was repeated carefully in full-screen to keep every field visible.

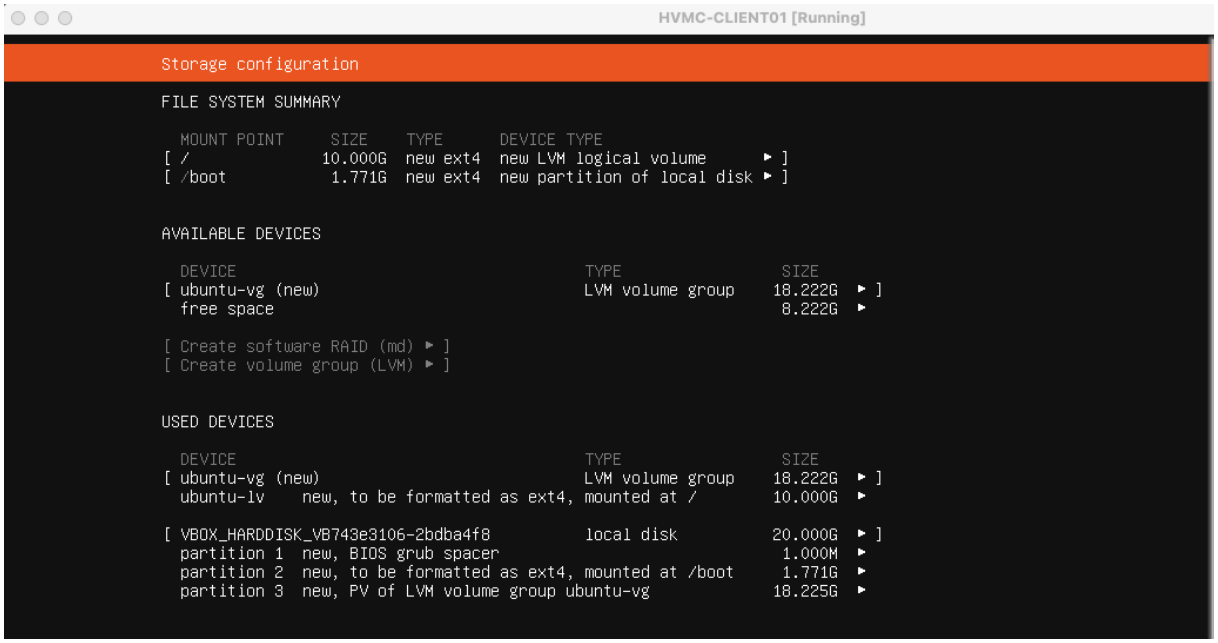


Figure 6: Storage summary: LVM, 10 GB root volume, free space left for later

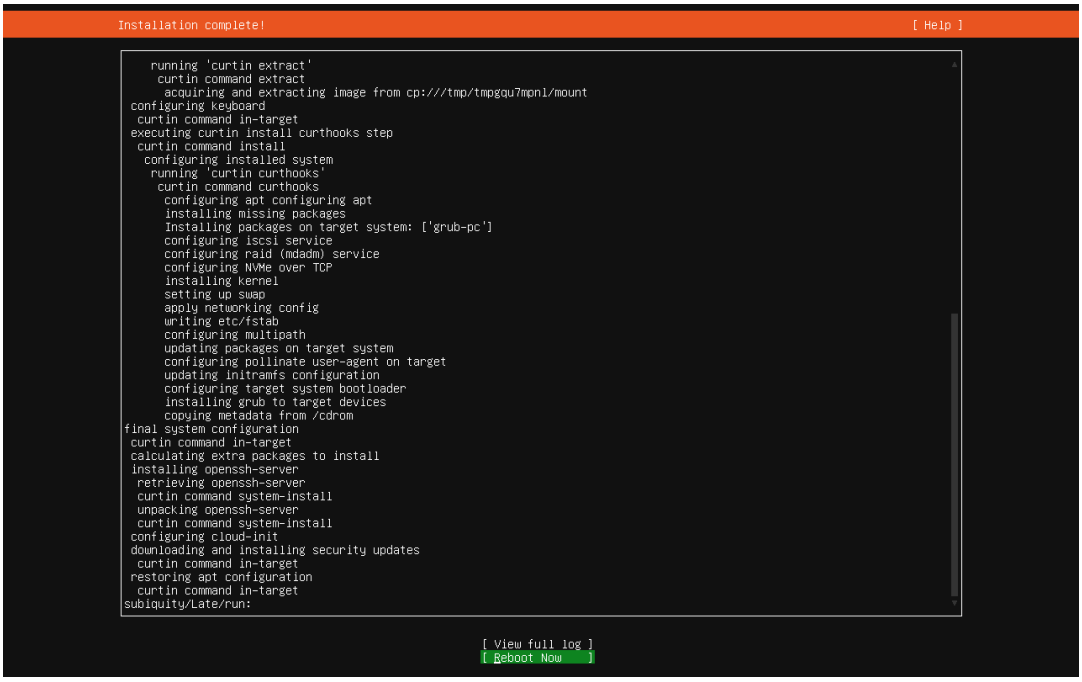


Figure 7: Installation complete; Reboot Now

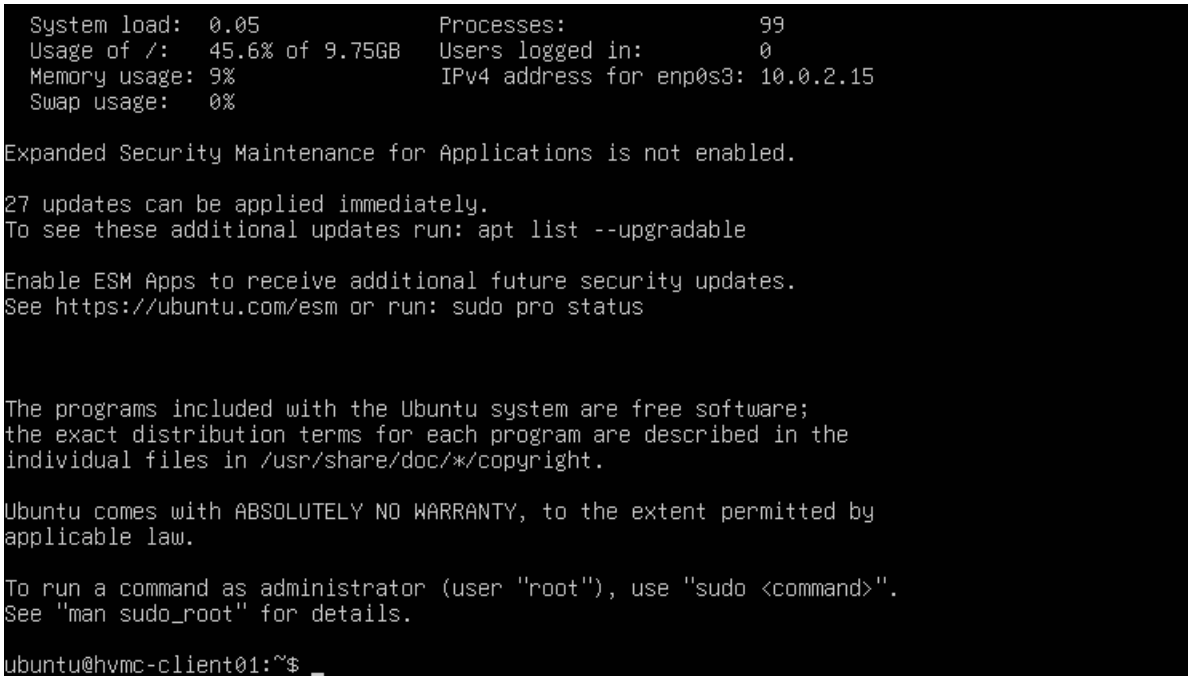


Figure 8: First login on the VM console after reboot

The administrator rights of the local account were tested before building anything on top of it:

```
sudo whoami
```

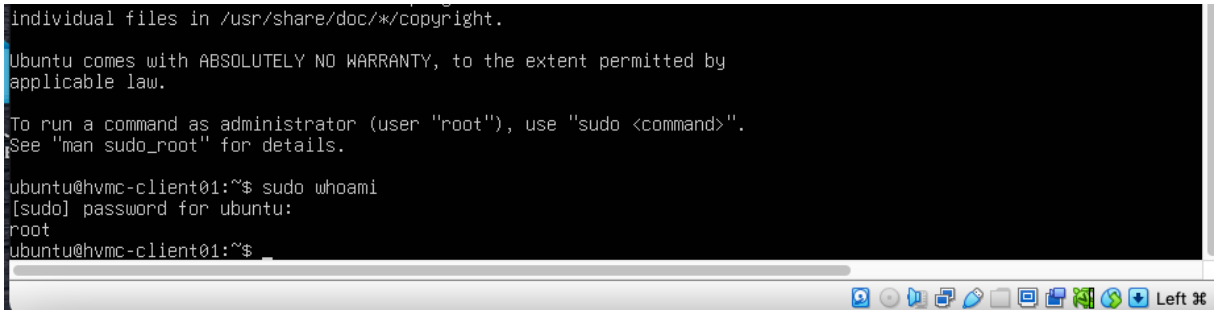


Figure 9: sudo whoami returns root: the local account and password work

5. Remote Access and Updates

The VM's addresses were read with `ip -4 -br a`. The lab adapter (enp0s8) received a temporary DHCP address, 192.168.56.4. From the Mac Terminal, SSH was used so commands can be copied and pasted. System updates were installed while the internet was reachable without the domain controller's DNS.

```
ip -4 -br a
ssh ubuntu@192.168.56.4
sudo apt update && sudo apt -y upgrade
```

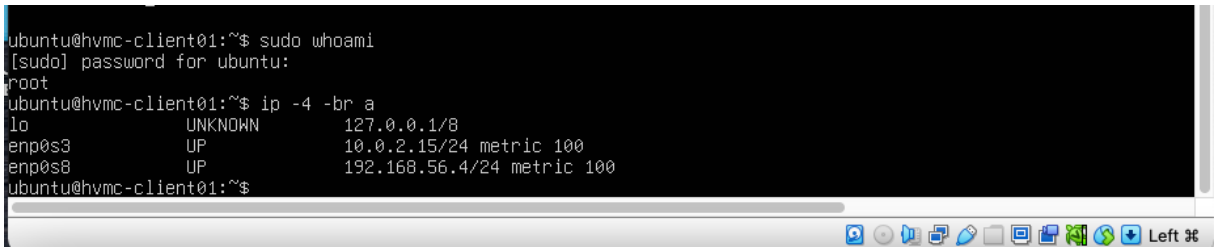


Figure 10: enp0s3 (NAT) 10.0.2.15 and enp0s8 (lab) 192.168.56.4

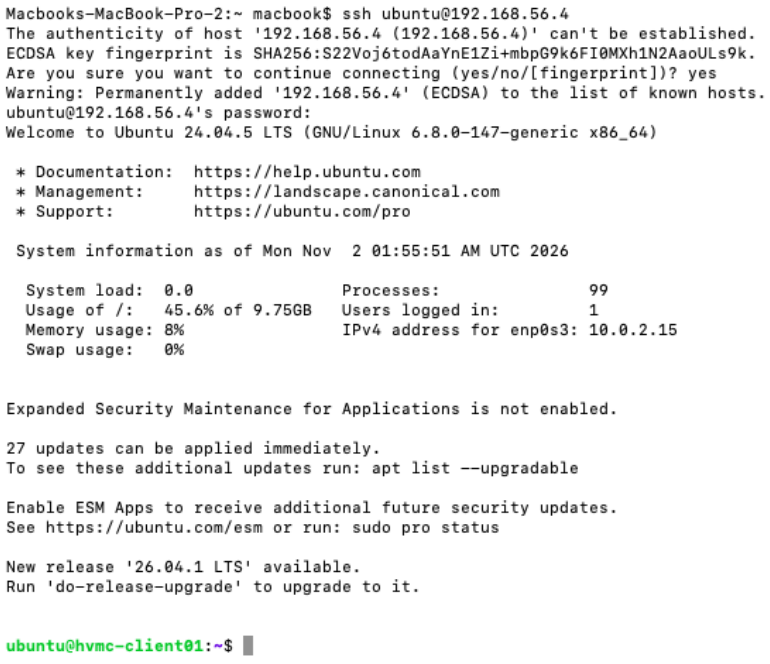


Figure 11: SSH login from macOS Terminal

6. Fixed Address, DNS and Snapshot 01

A domain member must keep a fixed address and use the domain controller for DNS. The netplan file was rewritten so that enp0s3 stays automatic but no longer supplies DNS, and enp0s8 uses **192.168.56.11** with the domain controller (192.168.56.10) as its DNS server and **hvmc.lan** as the search domain. Cloud-init's network control was disabled so the setting survives a reboot.

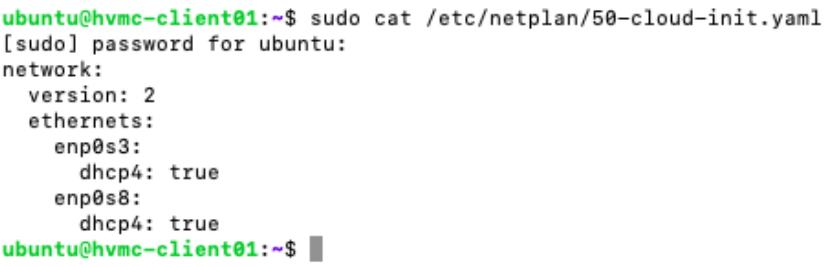


Figure 12: Original netplan: both adapters on DHCP

```
sudo tee /etc/netplan/50-cloud-init.yaml > /dev/null << 'EOF'
```

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp4-overrides:
        use-dns: false
EOF
sudo tee -a /etc/netplan/50-cloud-init.yaml > /dev/null << 'EOF'
    enp0s8:
      dhcp4: false
      addresses: [192.168.56.11/24]
      nameservers:
        addresses: [192.168.56.10]
        search: [hvmc.lan]
EOF
```

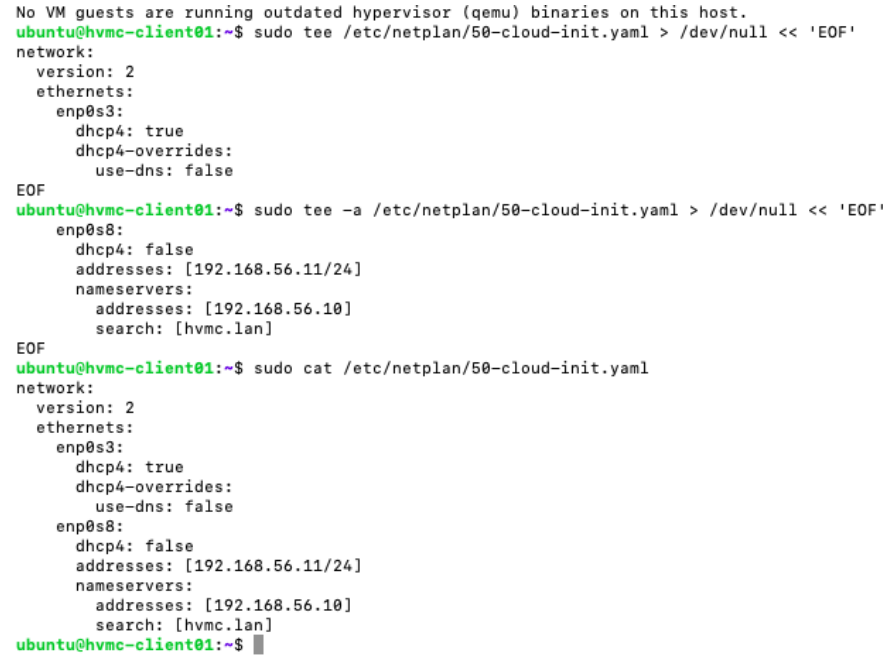


Figure 13: New netplan written and checked

```
echo 'network: {config: disabled}' | sudo tee /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
sudo chmod 600 /etc/netplan/50-cloud-init.yaml
sudo netplan apply
ssh ubuntu@192.168.56.11
```

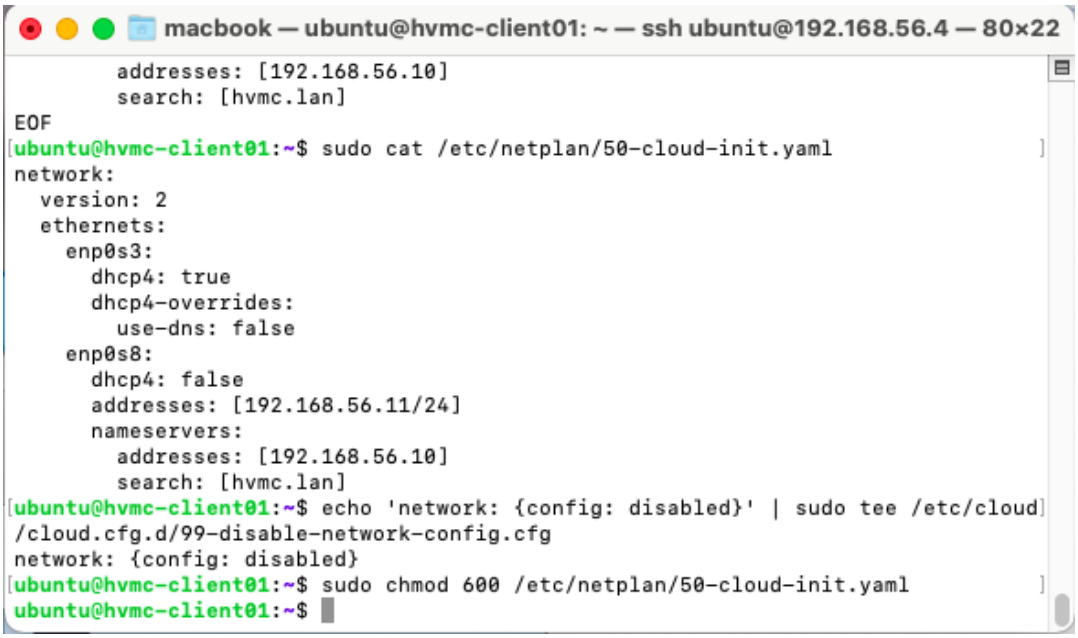


Figure 14: Cloud-init network control disabled; file permissions set

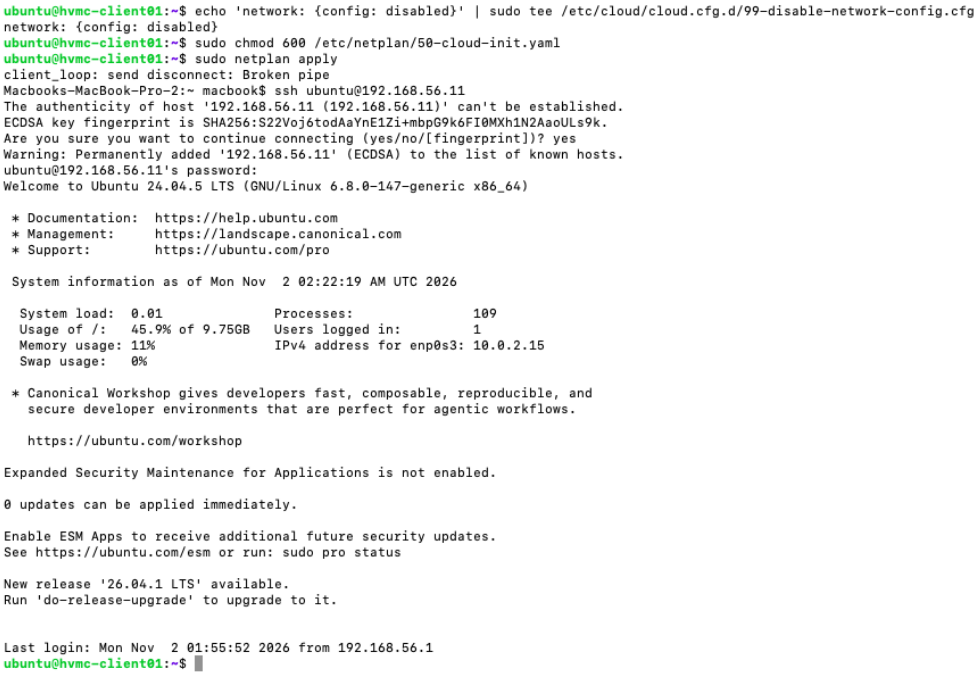


Figure 15: After netplan apply, reconnected at the permanent address 192.168.56.11

The client was then powered off and **snapshot 01 – “clean Ubuntu + fixed IP”** was taken.

```
sudo poweroff
```

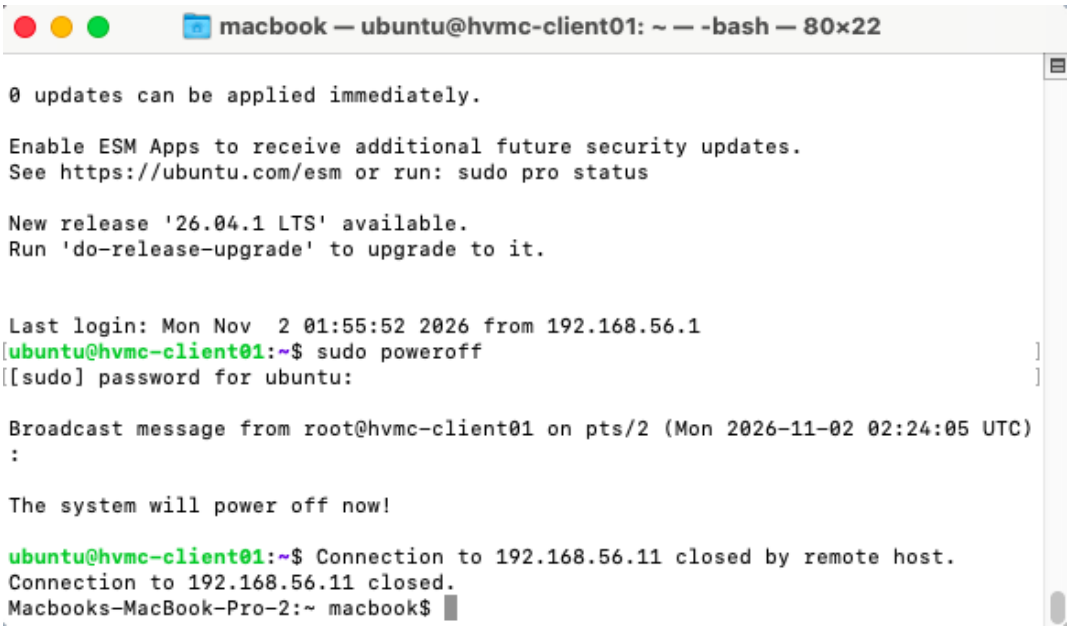


Figure 16: Clean shutdown before the snapshot

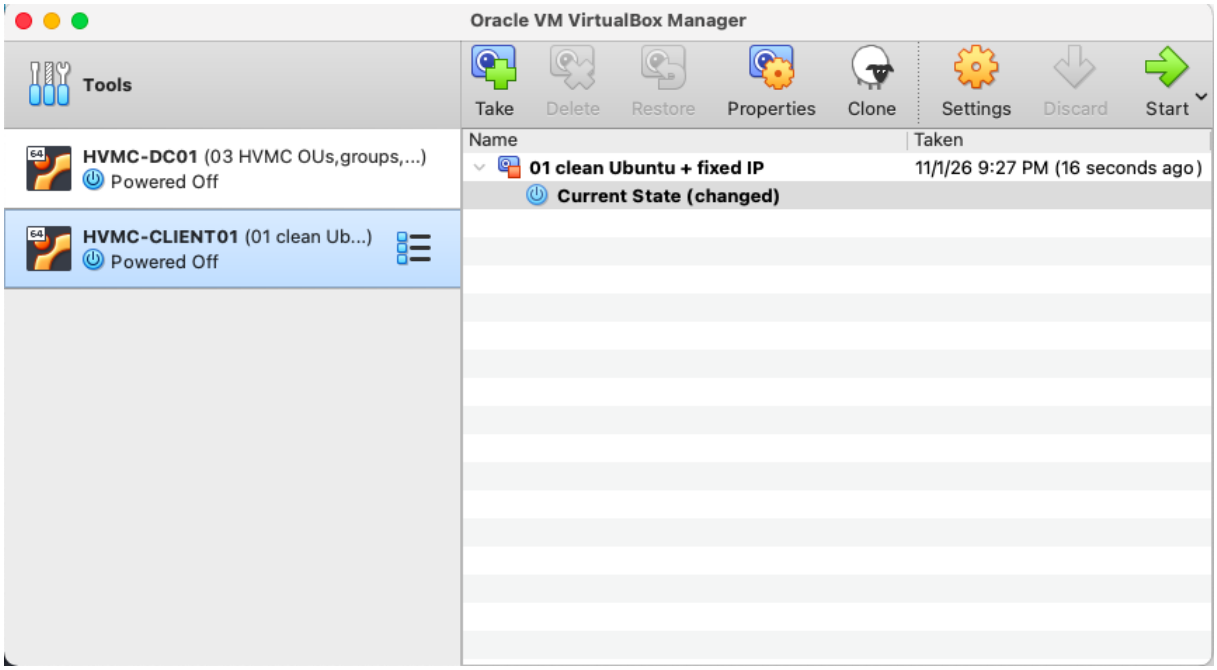


Figure 17: Snapshot 01 saved for HVMC-CLIENT01

7. Joining the Domain

7.1 Start the domain controller and test DNS

The domain controller was started with no window (Headless Start), then the client was started the same way. From the client, DNS was tested: the controller's name resolved to 192.168.56.10, which shows the client can find the domain.

```
host -t A hvmc-dc01.hvmc.lan
```

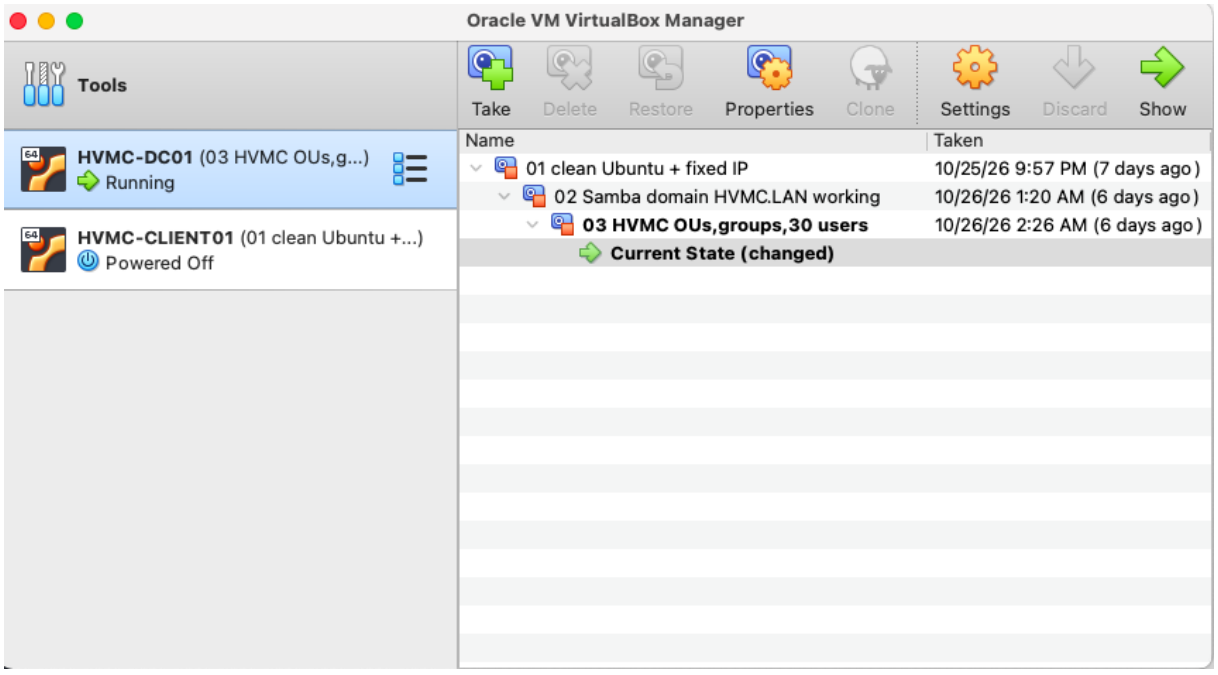


Figure 18: HVMC-DC01 running headless

```
Run 'do-release-upgrade' to upgrade to it.


Last login: Mon Nov  2 02:22:20 2026 from 192.168.56.1
ubuntu@hvmc-client01:~$ host -t A hvmc-dc01.hvmc.lan
hvmc-dc01.hvmc.lan has address 192.168.56.10
ubuntu@hvmc-client01:~$
```

Figure 19: hvmc-dc01.hvmc.lan resolves to 192.168.56.10

7.2 Install the join tools

realmd performs the join, adcli adds the computer to Active Directory, sssd and the libnss/libpam packages let domain users sign in, krb5-user provides Kerberos tools, and oddjob-mkhomedir creates home folders.

```
sudo DEBIAN_FRONTEND=noninteractive apt install -y realmd sssd sssd-tools \
  libnss-sss libpam-sss adcli samba-common-bin oddjob oddjob-mkhomedir \
  packagekit krb5-user
```

```
Could not execute systemctl:  at /usr/bin/deb-systemd-invoke line 148.
Setting up sssd-proxy (2.9.4-1.1ubuntu6.8) ...
Setting up sssd-dbus (2.9.4-1.1ubuntu6.8) ...
sssd-ifp.service is a disabled or a static unit, not starting it.
Setting up sssd-ad-common (2.9.4-1.1ubuntu6.8) ...
Created symlink /etc/systemd/system/sssd.service.wants/sssd-pac.socket → /usr/l
sssd-pac.service is a disabled or a static unit, not starting it.
Could not execute systemctl:  at /usr/bin/deb-systemd-invoke line 148.
Setting up sssd-krb5-common (2.9.4-1.1ubuntu6.8) ...
Setting up samba-common-bin (2:4.19.5+dfsg-4ubuntu9.8) ...
Setting up sssd-krb5 (2.9.4-1.1ubuntu6.8) ...
Setting up sssd-ldap (2.9.4-1.1ubuntu6.8) ...
Setting up sssd-ad (2.9.4-1.1ubuntu6.8) ...
Setting up sssd-tools (2.9.4-1.1ubuntu6.8) ...
Setting up sssd-ipa (2.9.4-1.1ubuntu6.8) ...
Setting up sssd (2.9.4-1.1ubuntu6.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
Processing triggers for oddjob (0.34.7-2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@hvmc-client01:~$
```

Figure 20: Join tools installed

7.3 Discover and join HVMC.LAN

`realm discover` confirms the client can see an Active Directory domain before anything is changed. The join then created the computer account in **OU=Computers**. The first attempt failed (most likely a mistyped password) and the second succeeded; `realm list` shows **configured: kerberos-member**.

```
realm discover hvmc.lan
sudo realm join --user=administrator \
  --computer-ou="OU=Computers,DC=hvmc,DC=lan" HVMC.LAN
realm list
```

```
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@hvmc-client01:~$ realm discover hvmc.lan
hvmc.lan
  type: kerberos
  realm-name: HVMC.LAN
  domain-name: hvmc.lan
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
ubuntu@hvmc-client01:~$
```

Figure 21: realm discover finds HVMC.LAN (server-software: active-directory)

```
  required-package: adcli
  required-package: samba-common-bin
ubuntu@hvmc-client01:~$ sudo realm join --user=administrator --computer-ou="OU=Computers,DC=hvmc,DC=lan" HVMC.LAN
[sudo] password for ubuntu:
Password for administrator:
See: journalctl REALMD_OPERATION=r1515.2439
realm: Couldn't join realm: Failed to join the domain
ubuntu@hvmc-client01:~$ sudo realm join --user=administrator --computer-ou="OU=Computers,DC=hvmc,DC=lan" HVMC.LAN
Password for administrator:
ubuntu@hvmc-client01:~$ realm list
hvmc.lan
  type: kerberos
  realm-name: HVMC.LAN
  domain-name: hvmc.lan
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@hvmc.lan
  login-policy: allow-realm-logins
ubuntu@hvmc-client01:~$
```

Figure 22: Join (first attempt failed, second succeeded) and realm list: kerberos-member

8. Domain User Sign-in

The client asked the domain about **sarah.chen**. The answer shows her domain user ID and her membership in **SG-Doctors-EHR-RW**, the group created from the Day 2 RBAC design. Automatic home folders were then enabled and Sarah signed in over SSH using her domain account.

```
id sarah.chen@hvmc.lan
sudo pam-auth-update --enable mkhomedir
ssh sarah.chen@hvmc.lan@192.168.56.11
whoami; hostname; pwd; id
exit
```

```
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@hvmc.lan
  login-policy: allow-realm-logins
ubuntu@hvmc-client01:~$ id sarah.chen@hvmc.lan
uid=842001113(sarah.chen@hvmc.lan) gid=842000513(domain users@hvmc.lan) groups=842000513(domain users@hvmc.lan),842001103(sg-doctors-ehr-rw@hvmc.lan)
ubuntu@hvmc-client01:~$
```

Figure 23: Domain lookup: sarah.chen with group sg-doctors-ehr-rw

```
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@hvmc.lan
  login-policy: allow-realm-logins
ubuntu@hvmc-client01:~$ id sarah.chen@hvmc.lan
uid=842001113(sarah.chen@hvmc.lan) gid=842000513(domain users@hvmc.lan) groups=842000513(domain users@hvmc.lan),842001103(sg-doctors-ehr-rw@hvmc.lan)
ubuntu@hvmc-client01:~$ sudo pam-auth-update --enable mkhomedir
ubuntu@hvmc-client01:~$
```

Figure 24: Home-folder creation enabled for domain users

```
For more details, please visit https://support.apple.com/kb/HT208050.
Macbooks-MacBook-Pro-2:~ macbook$ ssh sarah.chen@hvmc.lan@192.168.56.11
sarah.chen@hvmc.lan@192.168.56.11's password:
Permission denied, please try again.
sarah.chen@hvmc.lan@192.168.56.11's password:
Creating directory '/home/sarah.chen@hvmc.lan'.
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)
```

Figure 25: Signed in as sarah.chen@hvmc.lan; home directory created on first login

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

sarah.chen@hvmc.lan@hvmc-client01:~$ whoami
sarah.chen@hvmc.lan
sarah.chen@hvmc.lan@hvmc-client01:~$ hostname
hvmc-client01
sarah.chen@hvmc.lan@hvmc-client01:~$ pwd
/home/sarah.chen@hvmc.lan
sarah.chen@hvmc.lan@hvmc-client01:~$ id
uid=842001113(sarah.chen@hvmc.lan) gid=842000513(domain users@hvmc.lan) groups=842000513(domain users@hvmc.lan),842001103(sg-doctors-ehr-rw@hvmc.lan)
sarah.chen@hvmc.lan@hvmc-client01:~$
```

Figure 26: whoami, hostname, pwd and id as sarah.chen on hvmc-client01

```
sarah.chen@hvmc.lan@hvmc-client01:~$ whoami
sarah.chen@hvmc.lan
sarah.chen@hvmc.lan@hvmc-client01:~$ hostname
hvmc-client01
sarah.chen@hvmc.lan@hvmc-client01:~$ pwd
/home/sarah.chen@hvmc.lan
sarah.chen@hvmc.lan@hvmc-client01:~$ id
uid=842001113(sarah.chen@hvmc.lan) gid=842000513(domain users@hvmc.lan) groups=842000513(domain use
sarah.chen@hvmc.lan@hvmc-client01:~$ exit
logout
Connection to 192.168.56.11 closed.
Macbooks-MacBook-Pro-2:~ macbook$
```

Figure 27: Signed out

The chain that made this work: **client → domain controller DNS → Kerberos → the OUs, users and groups built in Week 1**. Sarah is an ordinary domain user, not an administrator, which is least privilege in practice.

9. Snapshot 02

After signing out and powering the client off, **snapshot 02 – "Joined HVMC.LAN, sarah.chen login works"** was taken as the restore point for the finished workstation.

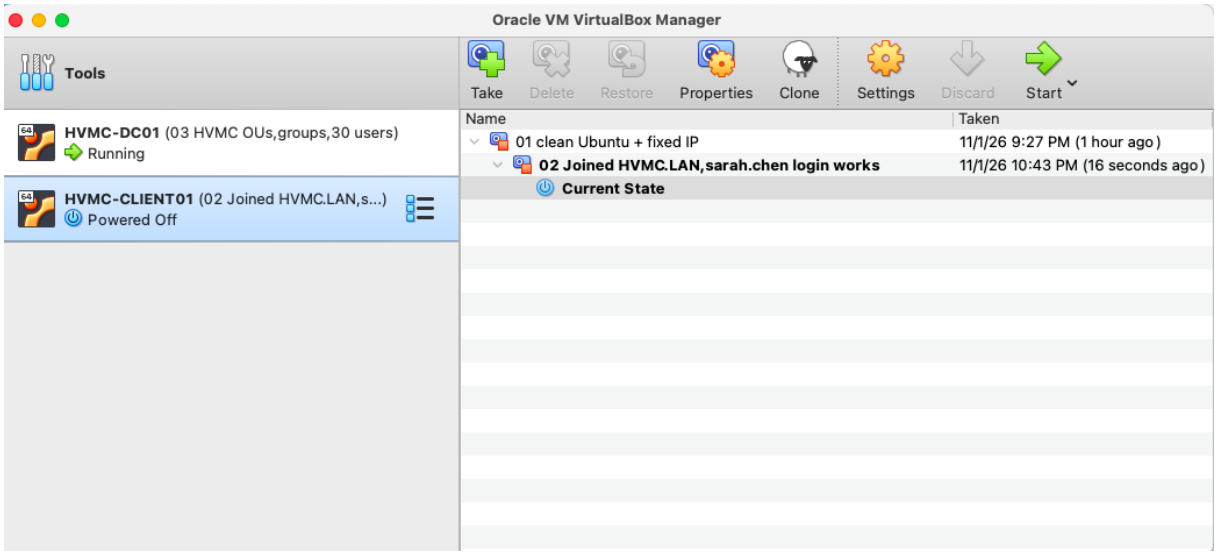


Figure 28: Snapshots 01 and 02 for HVMC-CLIENT01

10. Troubleshooting Log

Issue	Cause	Resolution
Pasted command ran as 00~ssh …	Terminal added stray characters when pasting	Typed the command by hand
SSH and sudo rejected the password after the first install	Saved password did not match what was expected; install window had been cut off	Deleted the VM (Remove → Delete all files) and reinstalled in full-screen
VM window not visible	Window hidden behind other windows	Used Show in VirtualBox and the Window menu
VirtualBox Manager quit unexpectedly	8 GB Mac under load	Reopened; the VM kept running
SSH to 192.168.56.11 timed out	The client VM was powered off	Started it with Headless Start and waited for boot
First realm join failed	Likely a mistyped administrator password	Ran the same command again; it succeeded

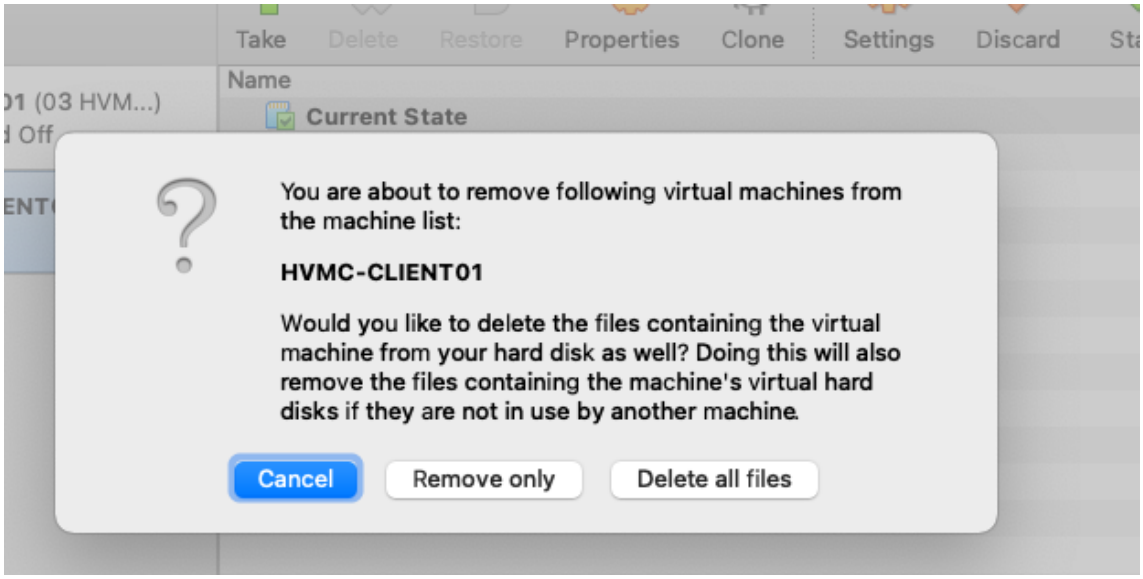


Figure 29: Removing the first client VM and deleting its files before the rebuild

11. Security Considerations

- All names, departments and users are **fictional**; no real patient or employee data is used.
- Passwords used in this lab are **practice-only** credentials for an isolated host-only network and are not used anywhere else.
- The client reaches the domain controller only over the private lab network.
- Domain users such as sarah.chen receive only the access of their group and no administrator rights (least privilege).

- Snapshots allow a quick rollback if a configuration change goes wrong.

12. Key Outcomes

- Built a second VM, HVMC-CLIENT01, with a fixed address (192.168.56.11) and DNS pointing at the domain controller.
- Joined it to HVMC.LAN and confirmed it is a Kerberos domain member in OU=Computers.
- Signed in as the domain user sarah.chen, showing that DNS, Kerberos, OUs, users and groups from Week 1 work end to end.
- Diagnosed and recovered from a password problem, a hidden window, a timeout and a failed first join.
- Saved two snapshots of the client.

13. Deliverables

- Week 2 – Client Workstation & Domain Join: Lab Documentation for GitHub Portfolio (this document).
- Client Computer Setup Record (steps with explanations and commands).
- VM snapshots 01–02 of HVMC-CLIENT01 (kept locally).